

WAFT: The Evolutionary Code Laboratory

SUMMARY

The Physics is the Scint System, which acts as a fitness function through Reality Fracture Detection.

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What is WAFT?

The Physics is the Scint System, which acts as a fitness function through Reality Fracture Detection. This system serves as natural selection that kills weak mutations. Agents face quests that test their ability to handle four types of errors: SYNTAX_TEAR for formatting errors, LOGIC_FRACTURE for math errors and contradictions, SAFETY_VOID for harmful content, and HALLUCINATION for fabricated facts. Agents must stabilize these errors to survive, and fitness is measured by stability, efficiency, and safety scores.

The Three Pillars

THE SUBSTRATE

WAFT is a Python framework for directed evolution of self-modifying AI agents. Think of it as an operating system for AI agent research projects. Instead of just building agents that execute code, WAFT enables you to breed agents that can modify their own code, evolve through mutations, and be tested in fitness systems with complete lineage tracking for scientific research.

How It Works

The Flight Recorder is a rigorous telemetry system for generating phylogenetic trees of agent lineage. Every evolutionary action is recorded with complete context including genome ID, parent ID, generation number, event type, payload with complete context, and fitness metrics. This enables reconstruction of complete family trees for scientific publication, allowing phylogenetic analysis, mutation impact measurement, fitness landscape mapping, and convergence analysis.

QUICK START

Install: `uv tool install waft`

Create: `waft new my_lab`

Verify: `waft verify`

WAFt is ambient, working quietly in the background without getting in your way. It is self-modifying, allowing projects to evolve their own structure over time. It is a meta-framework that orchestrates existing tools rather than replacing them. Everything is file-based, making it git-friendly and portable. The system includes gamification with D&D-style progression, epistemic tracking to know what you know and don't know, and scientific observation with complete lineage tracking.

Key Features

- | **Self-Modification:** Agents can modify their own code and evolve
- | **Fitness Testing:** Scint Gym evaluates agents through error handling
- | **Lineage Tracking:** Complete phylogenetic trees for scientific analysis
- | **File-Based:** Everything is plain text, git-friendly, no database
- | **Scientific Instrument:** Produces data for research on AI cognition